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Original Report

Measurable improvement in patient safety culture: A departmental experience with incident learning

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Abstract

Purpose: Rigorous use of departmental incident learning is integral to improving patient safety and quality of care. The goal of this study was to quantify the impact of a high-volume, departmental incident learning system on patient safety culture.

Methods and materials: A prospective, voluntary, electronic incident learning system was implemented in February 2012 with the intent of tracking near-miss/no-harm incidents. All incident reports were reviewed weekly by a multiprofessional team with regular department-wide feedback. Patient safety culture was measured at baseline with validated patient safety culture survey questions. A repeat survey was conducted after 1 and 2 years of departmental incident learning. Proportional changes were compared by χ^2 or Fisher exact test, where appropriate.

Results: Between 2012 and 2014, a total of 1897 error/near-miss incidents were reported, representing an average of 1 near-miss report per patient treated. Reports were filed by a cross section of staff, with the majority of incidents reported by therapists, dosimetrists, and physicists. Survey response rates at baseline and 1 and 2 years were 78%, 80%, and 80%, respectively. Statistically significant and sustained improvements were noted in several safety metrics, including belief that the department was openly discussing ways to improve safety, the sense that reports were being used for safety improvement, and the sense that changes were being evaluated for effectiveness. None of the surveyed dimensions of patient safety culture worsened. Fewer punitive concerns were noted, with statistically significant decreases in the worry of embarrassment in front of colleagues and fear of getting colleagues in trouble.

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Conclusions: A comprehensive incident learning system can identify many areas for improvement and is associated with significant and sustained improvements in patient safety culture. These data provide valuable guidance as incident learning systems become more widely used in radiation oncology.

Introduction

In medicine, a "culture of safety" refers to the collective values, attitudes, perceptions, competencies, and patterns of behavior that demonstrate a commitment to health and safety management.[1] When a positive culture of safety exists, a nonhierarchical, collaborative environment fosters openness, mutual respect, participation, and responsibility.[2,3] These cultural aspects of safety are associated with fewer adverse medical events.[4]

One tool with the potential to improve safety culture is incident learning, which refers to the feedback loop that begins at reporting of an event, followed by detailed analysis of contributing factors and culminating in the development of interventions to prevent recurrence.[5,6] In other high-risk sectors, incident learning is effective in reducing errors.[6,7] With regard to the intricate process of safely delivering therapeutic radiation, reliance on generic hospital error-reporting systems for quality improvement is limited by its reactive approach to errors after they have occurred, often without sufficient detail to identify contributing factors within the complex radiation oncology workflow. Additionally, "near-miss" incidents that never result in patient harm occur much more frequently than serious errors[8] and typically are underrepresented in such systems. Quality improvement is a perpetual effort, and both errors and near-miss incidents provide a vital learning tool to improve patient safety.[7,9,10] Recognizing this, reports at the professional society level advocate for the use of near-miss incident learning systems with formalized review processes,[11,12] and the national Radiation Oncology Incident Learning System has been developed.

In the absence of a positive patient safety culture, a significant proportion of errors or near-misses may go unreported. Suboptimal reporting and lack of engagement limit input into incident learning systems, and potential opportunities to further improve patient safety may go unrecognized. The use of an incident learning system itself may also exert changes on patient safety culture.[13] To the best of our knowledge, however, the link between incident learning and safety culture has not been evaluated in a quantitative manner. Using a validated survey instrument, we evaluated patient safety culture in our radiation oncology department and quantitatively assessed changes over a 2-year period after implementation of a high-volume, prospective incident learning program.

Methods and materials

This project received approval from the institutional review board at the University of Washington.

Incident reporting system

In February 2012, our department implemented a prospective, voluntary, Internet-based incident learning system based on existing incident reporting and learning design principles.[14-17] Our radiation oncology department has 2 computed tomography simulators, 4 linear accelerators, and specialized capabilities such as a fast neutron cyclotron, brachytherapy, and intraoperative radiation. Thirteen radiation oncologists treat an average of 1000 patients per year, with additional department members including 10 physicists, 10 radiation oncology residents, 2 medical physics residents, 5 dosimetrists, 20 radiation therapists, 2 advanced registered nurse practitioners, 8 registered nurses, and 8 clinical administrative support staff.

We adopted terminology consistent with that described in the literature for incident learning.[5] An incident is any undesired or unexpected event or condition that deviates from normal expectations that causes or has the potential to cause an adverse effect. A mistake is considered implementation of a plan unlikely to achieve its intended outcome even if executed correctly. A near-miss is an event or situation that could have resulted in an accident, injury, or illness but did not, either by chance or through intervention. Finally, an error is the failure to complete a planned action as intended or the use of an incorrect plan of action to achieve a given aim.

A full description of the incident reporting system will be published separately. Briefly, incidents are reported using a simple desktop interface available on all department computers. Reports contain the reporter's identification and a brief text description of the incident with automatic timestamp and numbering. An option for anonymity does exist if desired by the reporter. Each week, a multiprofessional committee consisting of physicians, physicists, dosimetrists, radiation oncology residents, medical physics residents, nurses, therapists, and administrative personnel review all reports and assign a near-miss severity index. If multiple reports are filed for the same event by the same user, only 1 report is counted. Each incident is discussed to identify possible corrective

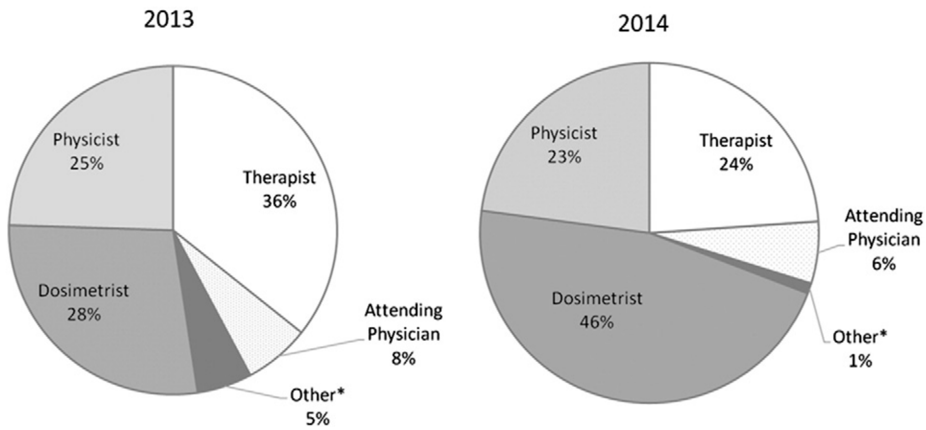

Figure 1 Distribution of incident reporters in 2013 and 2014. *Includes resident physicians, medical physics residents, advanced registered nurse practitioners, registered nurses, and administrative staff.

actions, and those that require further detailed investigation are subjected to in-depth root-cause analysis.[18] Feedback on workflow and policy changes that occur in response to incident reports is provided to the entire department at monthly department-wide quality improvement meetings. Additionally, a "good catch" system exists to recognize individuals who identify incidents that may not have been recognized by existing quality assurance measures.

Patient safety culture survey tool

Patient safety culture was evaluated by anonymous, online survey with content based on the Agency for Healthcare Research and Quality Hospital Survey on Patient Safety Culture and other surveys measuring safety culture.[19-21] Five general survey areas included (1) patient safety perceptions, (2) teamwork, (3) communication and punitive concerns, (4) responsibility and efficacy, and (5) feedback. Survey questions primarily use 5-point Likert scale responses. For questions that are negatively worded, a response of "strongly disagree" or "disagree" is interpreted as a positive response as described by the Agency for Healthcare Research and Quality.[19] For example, disagreement with the statement, "We work in 'crisis mode,' trying to do too much, too quickly," is considered a positive response, whereas agreement with this negatively worded statement indicates a negative response. Common perceived barriers to incident reporting are presented, with respondents asked whether these factors come to mind when they consider to report incidents. Finally, respondents were presented the opportunity to submit comments and perceptions about patient safety in free text.

The complete survey is provided in the online supplemental materials (available online only at www.practicalradonc.org). All department employees were first solicited to complete the survey online before implementation of the incident learning system in February 2012. Repeat survey was performed after 1 and 2 years of incident reporting in February 2013 and 2014, respectively.

Data analysis

Descriptive statistics of incident reporting system use are presented. Changes in percent positive response for each question item at 1 and 2 years of incident reporting were compared with the baseline survey response before incident reporting was initiated. Chi-square or Fisher exact tests were used where appropriate. All statistical tests were performed at a predefined level of significance of $P \leq .05$ using baseline results as the comparison group.

Results

Incident reporting

Between February 2012 and February 2014, a total of 1897 incident reports were filed (2012-2013, 836 reports; 2013-2014,1061 reports), representing on average 1.0 reports per patient treated. The majority of incident reports were filed by members of the therapist, dosimetrist, and physicist teams, with an increase in the proportion of reports filed by dosimetrists by 2 years (Fig 1). The distribution of reporters did vary significantly from month to month, although physicians consistently accounted for a minority of reports. In 2012, 57% of respondents indicated that they had not reported any incidents in the prior 12 months, with this number decreasing to 32% by 2013 (P = .02) and 29% in 2014 (P < .01).

Patient safety culture survey

The initial patient safety culture survey was performed in 2012, with responses received from 68 of 87 department

Table 1 Patient safety survey responses 2012-2014

	% Positive 2012 (n = 68)	% Positive 2013 (n = 70)	*P* value (ref 2012)	% Positive 2014 (n = 70)	*P* value (ref 2012)
Patient safety perceptions					
Errors/near-misses happen, but my team is so careful we don't have events to report [b]	49	69	.02 [a]	73	<.01 [a]
We have patient safety problems in this department [b]	47	63	.06	59	.18
It is just by chance that more serious mistakes don't happen around here [b]	59	67	.38	69	.29
Patient safety is never sacrificed to get more work done	59	50	.31	53	.50
I believe my colleagues value error and near-miss reporting	81	86	.50	79	.83
Teamwork					
We have enough staff to handle the workload	74	65	.28	57	.05 [a]
We use more agency/temporary staff than is best for patient care [b]	69	78	.26	57	.22
When one area in this unit gets really busy, others help out	55	57	.87	61	.49
When a lot of work needs to get done quickly, we work together as a team	88	86	.80	80	.25
In this department, people treat each other with respect	71	74	.70	73	.85
People support one another in this department	81	77	.68	79	.83
We work in "crisis mode", trying to do too much, too quickly [b]	31	44	.12	33	.86
When pressure builds up, my supervisor wants us to work faster, even if it means taking shortcuts [b]	68	81	.08	66	.86
Staff in this unit work longer hours than is best for patient care [b]	46	60	.09	44	1.0
Open communication and punitive concerns					
In this unit, we discuss ways to prevent errors from happening again	66	81	.05 [a]	86	.01 [a]
I'd be more likely to report errors/near-misses if it were anonymous [b]	38	60	.01 [a]	56	.04 [a]
Staff are afraid to ask questions when something does not seem right [b]	51	66	.12	67	.08
My colleagues would report an error or a near-miss that they caused	63	79	.06	77	.09
Staff worry that mistakes they make are kept in their personnel file [b]	29	44	.08	41	.16
My colleagues would report an error or a near-miss that I caused	84	87	.63	91	.20
Staff feel like their mistakes are held against them [b]	44	51	.40	56	.23
Staff feel free to question decisions/actions of those with more authority	35	41	.49	46	.30
When an event is reported, it feels like the person is being written up, not the problem [b]	47	61	.12	57	.31
Staff freely speak up if seeing something that may negatively affect patient care	69	76	.45	76	.45
Responsibility and efficacy					
I know how to report errors/near-misses within my department	78	90	.06	94	<.01 [a]
I know what kinds of errors/near-misses should be reported to my department	75	86	.13	93	<.01 [a]
I'd report errors/near-misses if I were not so busy [b]	63	59	.60	43	.02 [a]
I'd be more likely to report errors/near-misses if it were easier to do so [b]	28	54	<.01 [a]	59	<.01 [a]
I have confidence that my reports get used to improve our system	53	74	<.01 [a]	76	<.01 [a]
We are actively doing things to improve patient safety	84	91	.14	93	.07
Our procedures and systems are good at preventing errors from happening	63	77	.06	73	.27

(Continued)

Table 1 (*continued*)

	% Positive 2012 (n = 68)	% Positive 2013 (n = 70)	*P* value (ref 2012)	% Positive 2014 (n = 70)	*P* value (ref 2012)
Mistakes have led to positive changes here	75	84	.10	81	.31
My supervisor overlooks patient safety problems that happen over and over [b]	79	79	.83	84	.51
My supervisor seriously considers suggestions for improving patient safety	76	83	.40	77	1.0
It is my responsibility to report errors/near-misses within my department	96	99	.36	96	1.0
Feedback					
After we make changes to improve safety, we evaluate their effectiveness	46	66	.03 [a]	64	.04 [a]
We are given feedback about changes put into place based on event reports	46	51	.50	57	.23
My supervisor compliments team when job is done according to safety procedures	65	76	.19	75	.59
We are informed about errors that happen in this unit	56	64	.39	60	.73
I'd be more likely to report errors/near-misses if I received feedback afterwards	57	50	.40	53	.61

[a] $P \leq .05$.
[b] For these negatively worded statements "strongly disagree" or "disagree" indicates a positive response.

members (78% response rate). On the follow-up surveys, 70 of 87 department members responded in both 2013 and 2014, for an 80% response rate in each year.

Patient safety perceptions

Before initiation of incident learning, 49% of respondents disagreed with the statement, "I know errors/near-misses happen, but my team is so careful we do not have events to report." (Disagreement with this statement, as with other negatively worded questions, indicates a positive response.) This proportion had increased significantly after 1 and 2 years of incident reporting to 69% ($P = .02$) and 72% ($P < .01$), respectively, which indicates a greater awareness that errors can occur on one's own team (Table 1). Nonsignificant improvements were observed with the admission that patient safety problems exist and that it is just by chance that more serious mistakes did not occur. There were no negative changes in patient safety perceptions noted. The majority of respondents indicated that they believed their colleagues value incident reporting, and this remained unchanged in follow-up (2012, 81%; 2013, 86%; 2014, 79%).

Teamwork

In 2012, 74% of respondents indicated they believed there was enough staff to handle the workload, which decreased to 57% in 2014 ($P = .05$; Table 1). Most respondents thought that teams worked together when addressing work that needed to be done quickly (2012, 88%; 2013, 86%; 2014, 80%) and that people treated others with respect and supported one another in the department (2012, 81%; 2013, 77%; 2014, 79%; Table 1), although there were no statistically significant changes in these beliefs over time.

Open communication and punitive concerns

Sustained significant improvements were observed with the belief that the department was discussing ways to prevent errors from happening again (2012, 66%; 2013, 81% [$P = .05$]; 2014, 86% [$P = .01$]). Additional improvements were observed with fewer respondents concerned about speaking up when something does not seem right and more respondents believing that their colleagues would report errors that they themselves caused, although these changes did not reach statistical significance (Table 1).

One survey question probed attitudes on anonymous reporting, with the understanding that a positive safety culture is reflected by staff not critically concerned that their reports were kept anonymous. Before incident reporting, 38% of respondents disagreed with the idea that anonymity would make them more likely to report incidents. By 2013, this had improved to 60%, which indicates that preserving anonymity would not influence their likelihood of reporting ($P = .01$), with sustained improvement of 56% in 2014 ($P = .04$). There was no change in concern that mistakes were kept in personnel files or were held against the reporter. Additionally, small but nonsignificant improvements were observed with the comfort of staff in questioning the decisions of those with more authority, speaking up if they saw something that might adversely affect patient care, and disagreeing with

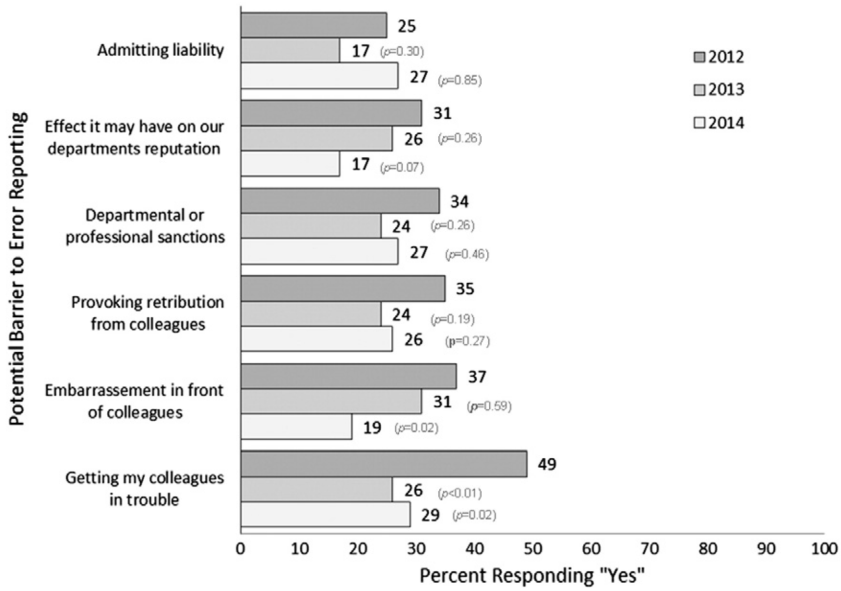

Figure 2 Self-reported perceived barriers to near-miss/error incident reporting between February 2012 and February 2014.

the notion that people were being written up as opposed to the event. We observed no evidence to suggest worsening of communication or punitive concerns.

Responsibility and efficacy

A number of statistically significant improvements in responsibility and self-efficacy (self-efficacy is the belief in one's own ability to impact change) were observed and are presented in Table 1. In 2012, 78% of respondents indicated that they knew how to report incidents in the department, with an increase to 94% by 2014 ($P < .01$). With 2 years of incident reporting experience, the proportion of respondents indicating knowledge of what type of errors/near-miss incidents should be reported increased from 75% to 93% ($P < .01$). The proportion of individuals reporting that they were confident that reports were being used to improve the system nearly doubled, from 28% in 2012 to 54% in 2013 ($P < .01$), with a continued increase to 59% in 2014 ($P < .01$). The majority of respondents believed that the department was actively trying to improve patient safety, had good procedures to prevent errors, and created positive change from mistakes, and these results remained unchanged in follow-up surveys (Table 1).

Feedback

A significant increase was observed in the number of respondents who believed that safety changes were evaluated for effectiveness, from 46% at baseline to 66% in 2013 ($P = .03$). This improvement was stable in 2014, with 64% agreeing with this sentiment ($P = .04$). Nonsignificant improvements were observed in the proportion of those who reported having received feedback on changes that resulted from reports, as well as having received compliments from senior members when established procedures were followed.

Barriers to reporting

A general pattern of decline in perceived barriers was noted for nearly all surveyed items. By 2014, only 19% of respondents reported concerns that incident reporting may cause embarrassment in front of colleagues, compared with 37% in 2012 ($P = .02$; Fig 2). A stable and sustained significant decrease in concern that error reporting would get colleagues in trouble was also observed (2012, 49%; 2013, 26% [$P = .01$]; 2014, 29% [$P = .02$]; Fig 2). Nonsignificant decreases were observed in the proportion of individuals who worried about the effect that reporting errors would have on the department's reputation, getting sanctioned by the department, and provoking retribution from colleagues. The one exception was that for admitting liability, there was an initial nonsignificant decline, with a return to a similar level before implementation of incident reporting (Fig 2).

Perceived sources of error

Self-reported common sources of error are presented in Fig 3. The most commonly perceived source of error in all survey periods was communication errors. Additional surveyed sources of error included failure to follow standardized procedures, technical failures, insufficient training, and too high a workload, which were indicated as "sometimes" being a contributor by the majority of respondents, without significant change over the study period.

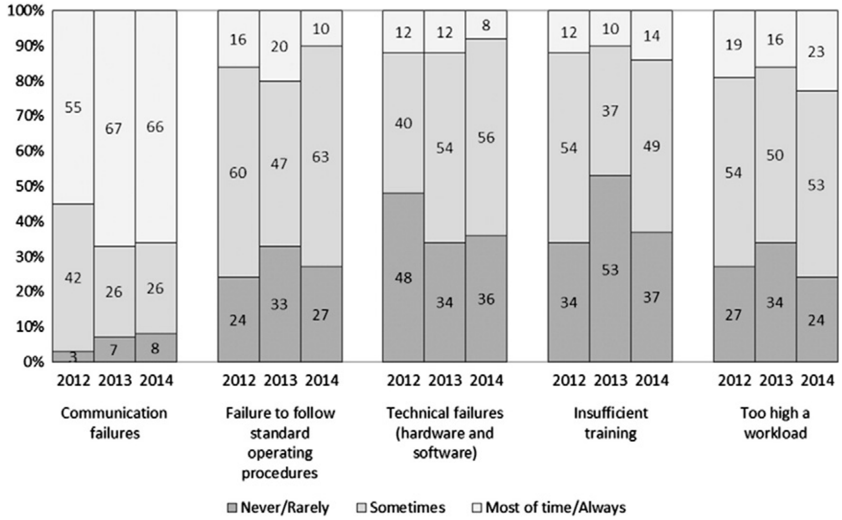

Figure 3 Self-reported frequency of common contributing factors to errors by year between February 2012 and February 2014.

Discussion

With 2 years of experience with incident learning, we have demonstrated measurable improvements in patient safety culture with enhanced communication, greater awareness of patient safety issues, improved self-efficacy, and fewer punitive concerns. The majority of these improvements were observed within the first year and continued to be statistically significant at 2 years, with additional areas of safety culture enhancement noted with longer use of incident learning. No decrement in patient safety culture was observed.

Our department implemented an incident reporting and learning system with a rapid increase in use and with the majority of incident reports filed by dosimetrists, therapists, and physicists. This balanced distribution of reports by professional teams differs from that seen in other studies[22,23] and may be due to the high levels of engagement by various teams, with incidents detected and reported all along the radiation oncology workflow. It is worth noting, however, that within professional teams, the level of participation in reporting incidents is variable, and at 2 years, 29% of respondents indicated that they had not filed an incident report within the past year. Cross-sectional data suggest that increased frequency of error reporting is associated with fewer adverse events.[4] During the first 2 years of incident reporting and learning, we have observed an increase in the number of reports with a concurrent decrease in the average near-miss severity index scores of reports (data not presented). With continued use, it may be that the number of reports will plateau and then decline, but currently this system continues to find more and more areas for minor improvement to prevent the rare major event, which is the foundation of near-miss incident learning systems. Further analysis of individual use of the incident reporting system and of the origin and detection of incidents within the radiation oncology workflow is ongoing.

In 2014, more respondents had concerns about having enough staff to handle the workload than in 2012. During this same period, there were no significant changes in reported feelings of trying to do too much, too quickly; being pressured to work faster; or working longer hours than was best for patient care. Although speculative, these data may suggest that with incident reporting experience and safety culture improvement, there is heightened awareness of the importance of adequate staffing to reduce and prevent errors and near-misses.

As an indicator of patient safety, improved safety culture is associated with fewer medical adverse events.[4,20] A means by which safety culture can positively impact patient safety is through greater engagement and participation in quality improvement initiatives such as incident learning systems, a proven tool in reducing error rates.[8,16] Fear of retribution or disciplinary action (punitive concerns), however, can impede the reporting of events, and it is recognized that a nonpunitive or "no-blame" climate is critical to the success of incident learning systems.[6,24,25] The surveyed items that focused on punitive concerns all demonstrated improvement without evidence of increased worry that mistakes were held against reporters or kept in personnel files. Survey items that examined perceived barriers to incident reporting, most of which were punitive in nature, similarly demonstrated improvement or stability. In an attempt to address punitive concerns, anonymity is used by many incident reporting systems.[7] Within our own incident reporting system, anonymity is optional, and we continue to demonstrate high use, with anonymous reports constituting only 2% of incident reports. This has greatly enhanced our ability to gather further details about reported events while simultaneously reinforcing a nonpunitive, no-blame culture.

At baseline, we observed a patient safety culture characterized by high levels of teamwork, mutual respect, and support. Although a good safety culture is important in adopting and accepting an incident learning system, there exists a reciprocal relationship, with widespread use of incident reporting and learning further enhancing patient safety culture, as seen in our results. In reviewing their institution's experience with incident learning, Clark et al[13] reported subjective improvements in engagement, empowerment, and self-efficacy in improving safety and quality of patient care. Here, we have demonstrated measureable improvements in awareness, participation, and confidence that individual efforts are contributing to safety improvement that provide quantitative support for these observations.

Clear departmental and institutional commitments with leadership involvement are key factors for incident learning to have a successful impact on patient care.[13,15,23] The described efforts require members of every team in the department to participate in the weekly multiprofessional review and further investigate details of reported incidents to inform corrective actions. In our experience, this investment has resulted in a shortened life cycle from event report to action while also maintaining engagement and providing prompt feedback. We believe these factors contribute to the observed patient safety culture improvements. In addition to enhancing patient safety culture and fostering improved interprofessional communication, our incident learning system has provided a valuable experience for radiation oncology and physics residents to learn about safety and quality improvement through longitudinal involvement during their training. Lastly, the prospective database has provided a searchable means by which issues can be tracked, trends can be observed, and associations with changes can be identified.

Strengths of this study include high patient safety survey response rates and the longitudinal component that measured safety culture before and after implementation of an incident learning system. There are several limitations to our study. There exists the possibility of detecting statistically significant associations by chance alone in the setting of multiple testing. However, we are reassured that the significant findings accurately reflect improvement in patient safety culture given the general trends of improvement observed with both 1 and 2 years of follow-up without evidence of decrement in safety culture as measured by survey items. Another limitation is the inability to definitively identify causality from the observed associations between increasing incident learning experience and patient safety culture. Finally, these findings may not be readily generalizable to other practice settings or geographies.

Conclusions

Implementation of near-miss incident reporting is associated with improved patient safety culture reflected by increased engagement, greater awareness of safety issues, decreased punitive concerns, and greater sense of self-efficacy in improving patient safety. As incident reporting and learning systems such as the national Radiation Oncology Incident Learning System become more widely used, the design factors of reporting systems and organizational culture will play a key role in patient safety efforts and perpetual quality improvement.

Acknowledgment

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